

Manuel Fernández López

Postdoctoral Researcher, Lattice Field Theory Group

📍 Hsinchu, Taiwan ✉️ manuel.fernandezl@nycu.edu.tw

Profile

- Condensed matter theorist specializing in topologically ordered, strongly correlated systems and their applications to quantum information science.

Research Focus

- Strongly correlated quantum materials
- Interaction-driven topological transitions
- Flat-band topological superconductivity in frustrated lattices
- Green function zeros to classify topological strongly correlated systems

Education

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| PhD | Autonomous University of Madrid (UAM), Physics <ul style="list-style-type: none"> • Thesis: "Fractionalized topological phases in strongly correlated materials with spin-orbit coupling" • Supervisor: Prof. Jaime Merino • Defense: Oct 2024 — Cum laude | Madrid, Spain
2019 – 2024 |
| MSc | Autonomous University of Madrid (UAM), Condensed Matter Physics and Biological Systems <ul style="list-style-type: none"> • Thesis: "Spin-orbit coupling and Coulomb interaction effects in decorated honeycomb lattices" | Madrid, Spain
2017 – 2018 |
| BSc | Autonomous University of Madrid (UAM), Physics <ul style="list-style-type: none"> • Theoretical final project: "Bosonic string theory for the derivation of the graviton and compactification of 25+1 dimensions." • Experimental final project: "Data analysis for the determination of the Z₀ boson mass using LHC data." • High honors in Thermodynamic and Statistical Physics I and II, Quantum Mechanics II, Physical Electronics, Magnetism and Superconductivity | Madrid, Spain
2013 – 2017 |

Experience

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| Institute of Physics, National Yang Ming Chiao Tung University, Postdoctoral Researcher, Lattice Field Theory Group <ul style="list-style-type: none"> • Member of the Lattice Field Theory Group at NYCU, translating computational techniques between condensed matter physics and non-perturbative QCD. • Applying parton-based and exact diagonalization methods developed for strongly correlated, topologically ordered systems to open questions in hadron structure. • Building complementary expertise in tensor network and DMRG approaches to lattice gauge theories. • Research project: Tensor networks and quantum computing for high-energy physics (NSTC grant 112-2112-M-A49-021-MY3; PI: Prof. C.-J. David Lin) — contributing to a project applying tensor-network and quantum-computing methods, developed for strongly correlated condensed matter systems, to problems in high-energy physics. | Hsinchu, Taiwan
Sept 2026 – present
1 month |
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Autonomous University of Madrid (UAM), Associate Researcher

- Developed theoretical approaches to compute Green's function zeros in topological Mott insulators; application to the Kitaev-Hubbard model (preprint in final stage).
- Proposed novel topological strongly correlated systems to apply AI-driven methodologies developed within the group.
- Prepared a referee report for Physical Review Letters; regularly handled editorial/referee correspondence as corresponding/co-author.
- Research project: New concepts and methods for topological strongly correlated quantum matter (Ref. PID2022-139995NB-I00; IP: Jorge Bravo Abad) — computed non-local Green's function in-gap zeros delineating a fractionalized protected state (spinon drumhead states), and led a research line computing Green's function zeros in Kitaev materials such as α -RuCl₃, opening a route to identify topological phases in correlated superconductors.
- Supervised a Master's student (March 2024 – March 2025) on nodal-line Mott insulator physics; acted as corresponding author on the resulting publication (J. C. Luaces, M. F. López, J. Bravo Abad & J. Merino, "Topological strongly correlated phases in orthorhombic diamond lattice compounds," Phys. Rev. B 111, 165133, 2025), guiding the full submission and peer-review process.
- Co-supervising a Ph.D. student (ongoing, through weekly group meetings) on inhomogeneous superconductivity in layered cuprates and nickelates, using extended Gutzwiller approximations complemented with AI-trained variational and numerical methods such as Monte Carlo; a collaborative manuscript is currently in its final stage.

Madrid, Spain
Nov 2024 – Aug 2026
1 year 10 months

Autonomous University of Madrid (UAM), Researcher Formation Fellow (FPI)

- Implemented a finite-T slave-rotor mean-field framework for topological Mott transitions; benchmarked local limits against DMFT.
- Identified flat-band topological superconductivity and localized Majorana edge modes in metal-organic frameworks (Phys. Rev. B 106, 235129, 2022).
- Performed slab Green's-function calculations revealing spinon Fermi arcs, a novel type of topological surface state (Phys. Rev. B 109, 085137, 2024).
- Derived effective Hubbard models from first-principles DFT band structures for strongly correlated-material modeling (Phys. Rev. B 111, 165133, 2025).
- Research project: New phenomena and applications of topological strongly correlated materials in and out of equilibrium (Ref. RTI2018-098452-B-I00; IP: Jaime Merino Troncoso, FPI PRE2019-087603) — supervised a master's student on code adaptation for correlated nodal-line semimetals, providing the first theoretical realization of the observed pressure-induced Mott transition; developed a semi-analytical technique extended to finite temperature revealing that non-local correlations lead to spinon Fermi arcs, with predicted ARPES signatures on pyrochlore iridates; investigated correlated topology in flat-band decorated honeycomb lattices motivated by MOFs, with exact diagonalization revealing a flat-band quantum spin liquid consistent with experiments; and explored flat-band unconventional superconductivity from doping this spin liquid, characterizing localized Majorana edge modes and their potential for topological quantum computing.
- Lecturer & Lab Supervisor, Physics Laboratory course for Environmental Science undergraduates, 2nd year (2021–2022).

Madrid, Spain
Sept 2019 – Oct 2024
5 years 2 months

Autonomous University of Madrid (UAM), Scientific Technician, National Research Project

- Adapted master's thesis results into a peer-reviewed publication and managed correspondence with editors and referees (Phys. Rev. B 100, 075154, 2019).
- Prepared departmental talks and presented results at a national conference.
- Research project: New electronic and photonic materials based on topological strongly correlated systems (Ref. MAT2015-66128-R; IP: Jaime Merino Troncoso) — implemented Hartree-Fock calculations in a decorated honeycomb lattice relevant to certain MOFs, identifying topological semimetals; presented findings at national conferences and acted as corresponding author in the peer-review process.

Madrid, Spain
Mar 2018 – Aug 2019
1 year 6 months

Journal Articles

- J. C. Luaces, M. F. López, J. Bravo Abad & J. Merino, "Topological strongly correlated phases in orthorhombic diamond lattice compounds", *Phys. Rev. B* **111**, 165133 (2025)
- M. F. López, I. García Elcano, J. Bravo Abad & J. Merino, "Emergence of spinon Fermi arcs in the Weyl-Mott metal insulator transition", *Phys. Rev. B* **109**, 085137 (2024)
- M. F. López & J. Merino, "Bad topological semimetals in layered honeycomb compounds", *Phys. Rev. B* **105**, 115138 (2022)
- M. F. López, B. J. Powell & J. Merino, "Topological superconductivity from doping a triplet quantum spin liquid in a flat band system", *Phys. Rev. B* **106**, 235129 (2022)
- J. Merino, M. F. López & B. J. Powell, "Unconventional superconductivity near a flat band in organic and organometallic materials", *Phys. Rev. B* **103**, 094517 (2021)
- M. F. López & J. Merino, "Magnetism and topological phases in an interacting decorated honeycomb lattice with spin-orbit coupling", *Phys. Rev. B* **102**, 035157 (2020)
- M. F. López & J. Merino, "From quantum anomalous Hall phases to topological metals in interacting decorated honeycomb lattices", *Phys. Rev. B* **100**, 075154 (2019)

Peer-Review Activity

- Referee report prepared for Physical Review Letters (submitted Aug 2025; editorial decision Sep 2025), assessing significance, methodological soundness and clarity of presentation.
- Regularly manage reviewer/editor correspondence for co-authored and first-authored manuscripts, including coordinating responses, revising manuscripts, and ensuring reporting/figures meet journal standards.
- Peer reviewer for Physical Review B, first as co-referee with supervisor, later as an independent referee.
- Experience assessing reproducibility, figure clarity, methodological soundness and experimental relevance in theoretical manuscripts, with a focus on transparent availability of data and custom code.

Presentations

- "From Quantum Anomalous Hall Phases to Topological Metals in Interacting Decorated Honeycomb Lattices" — Oral contribution, INC Young Researchers Meeting, Madrid (2019)
- "Magnetism and Topological Phases in an Interacting Decorated Honeycomb Lattice with Spin-Orbit Coupling" — Oral contribution, INC Young Researchers Meeting, Madrid (2021)
- "Topological Phases Induced by Interactions and Spin-Orbit Coupling in Decorated Honeycomb Lattices" — Oral contribution, APS March Meeting, online (2021)
- "Bad Topological Semimetals in Layered Honeycomb Compounds" — Oral contribution, APS March Meeting, Chicago, IL, USA (2022)
- "Bad Topological Semimetals in Layered Honeycomb Compounds" — Invited talk, School of Mathematics and Physics, University of Queensland, Australia (2022)
- "Emergence of Spinon Fermi Arcs in the Weyl-Mott Metal-Insulator Transition" — Oral contribution, INC Young Researchers Meeting, Madrid (2024)
- "Bad Weyl Semimetals and Spinon Fermi Arcs in a Model for Pyrochlore Iridates" — Oral contribution, IOP CMQM 2024, St. Andrews, Scotland (2024)

Technical Skills

- Semianalytical / Numerical: Parton mean-field theories, exact diagonalization, solid working knowledge of (cluster) DMFT and QMC methods, topological invariants calculation in the absence of Landau quasiparticles
- Programming & Software: LaTeX, Fortran, Matlab (high proficiency); Adobe Suite, Wolfram Mathematica (advanced); Python (beginner); Windows, macOS, Linux
- Communication: Manuscript evaluation, referee correspondence, figure/table preparation
- Languages: English (fluent), Spanish (native)

Honors & Fellowships

- FPI Research Fellowship, Autonomous University of Madrid (UAM) (2019–2024)